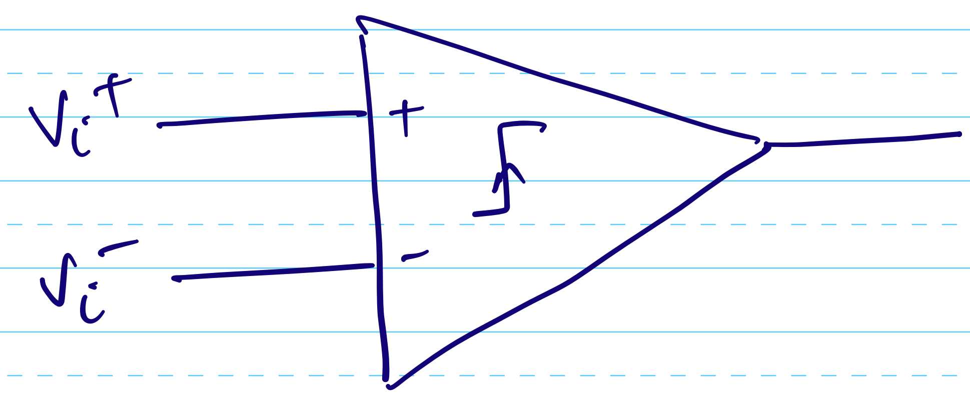
Session 7B

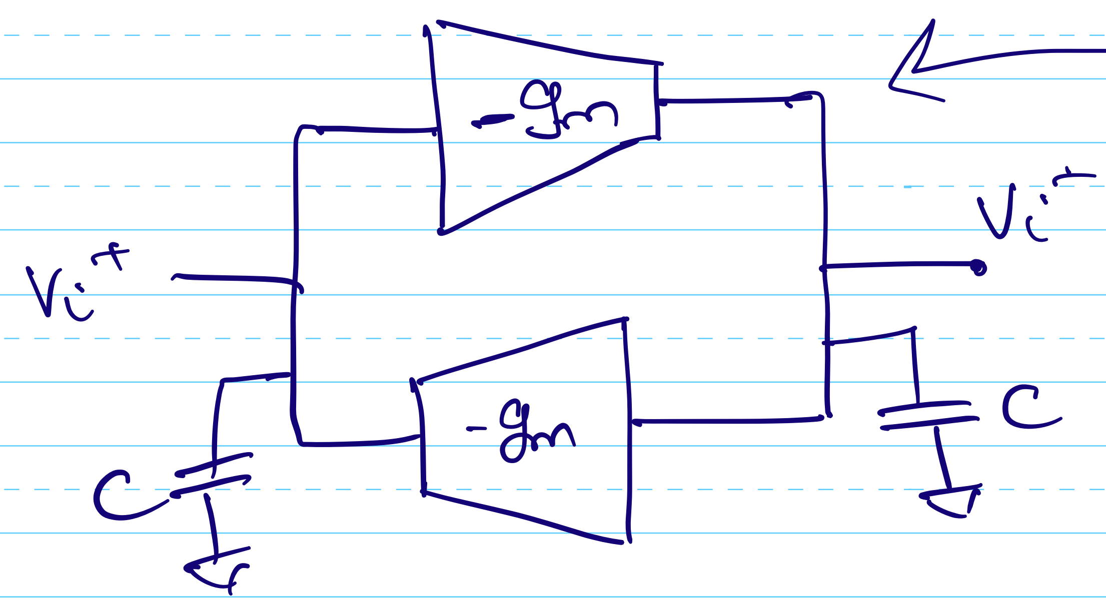
Comparators

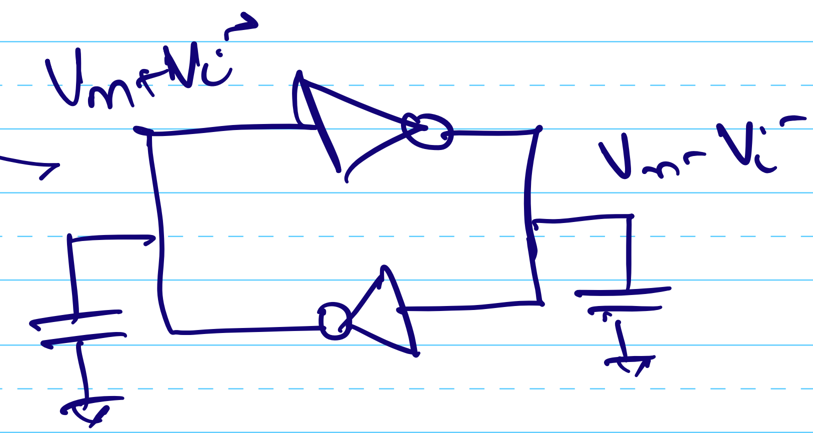


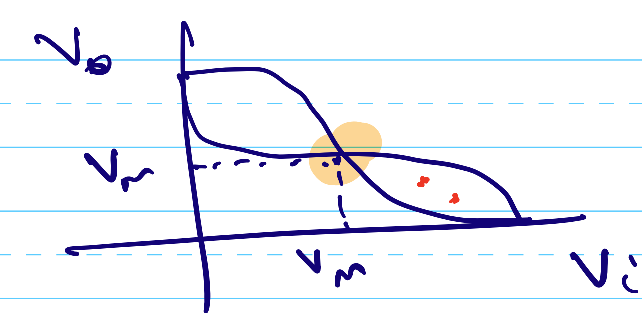
0 if $V_i^+ < V_i^-$ $\rightarrow$ 0V

1 if $V_i^+ > V_i^-$ $\rightarrow$ V_{DD}

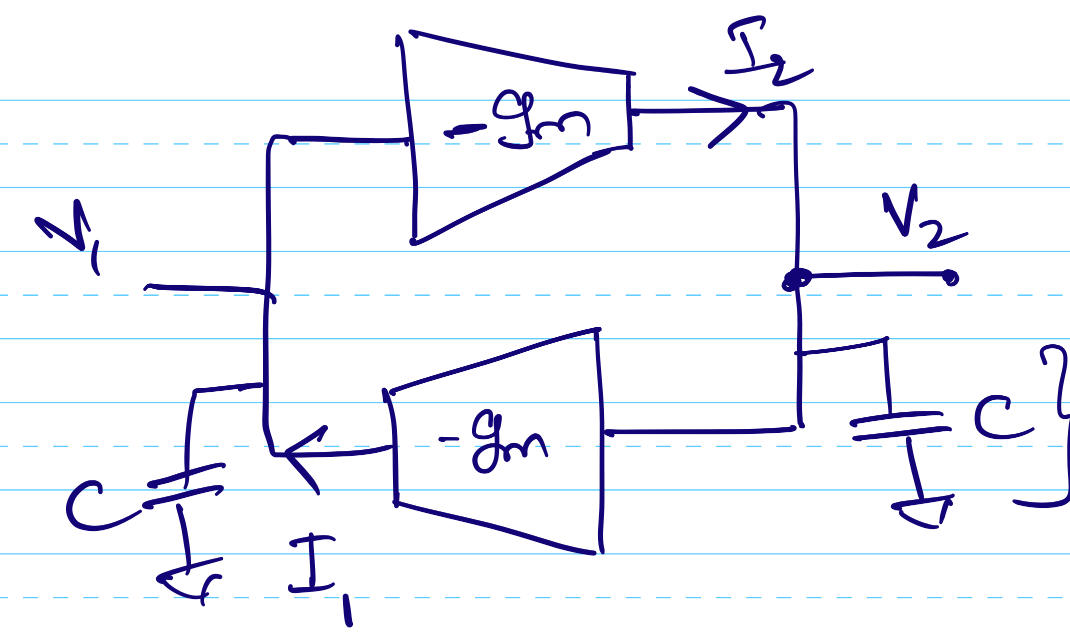
Positive feed back







$$I_1 = -g_m V_2$$



$$I_2 = -g_m V_1$$

$$C\frac{dV_2}{dt} = I_2 \qquad C\frac{dV_1}{dt} = I_1$$

$$C\frac{dV_2}{dt} = -g_m V_1 \quad \text{— A} \qquad C\frac{dV_1}{dt} = -g_m V_2 \quad \text{— B}$$

$$C\frac{dV_2}{dt} - C\frac{dV_1}{dt} = -g_m V_1 + g_m V_2$$

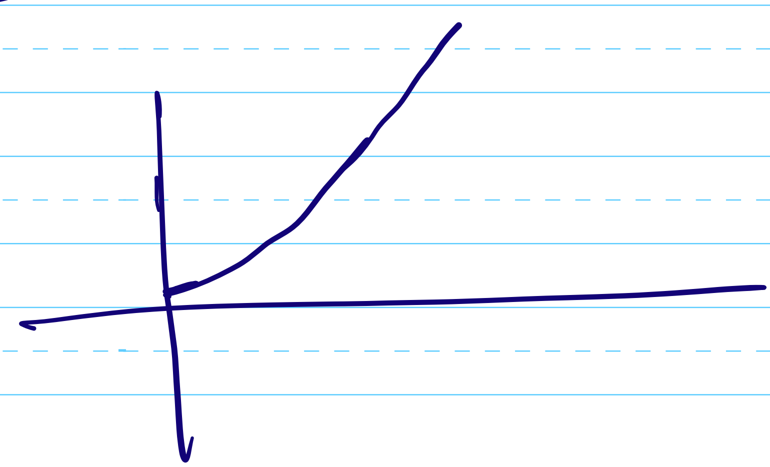
$$C\frac{dV_2}{dt} - C\frac{dV_1}{dt} = -g_m V_1 + g_m V_2$$

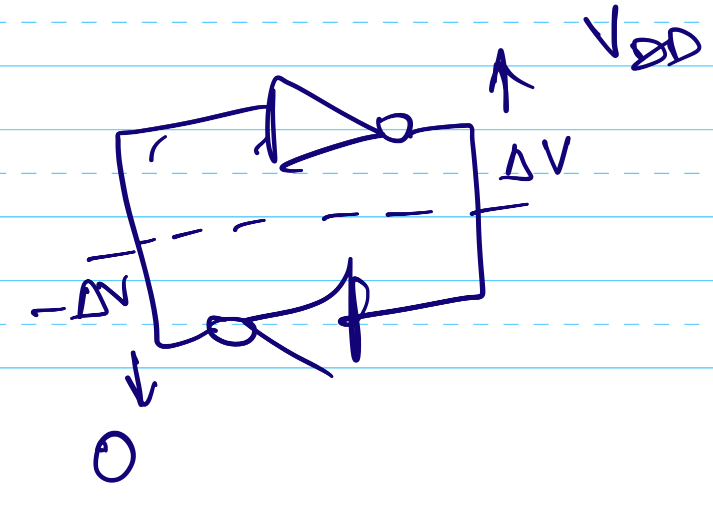
$$C\frac{d}{dt}(V_2 - V_1) = g_m (V_2 - V_1) \qquad \Delta V = V_2 - V_1$$

$$\int \frac{d\,\Delta V}{\Delta V} = \int \frac{g_m}{C}\, dt$$

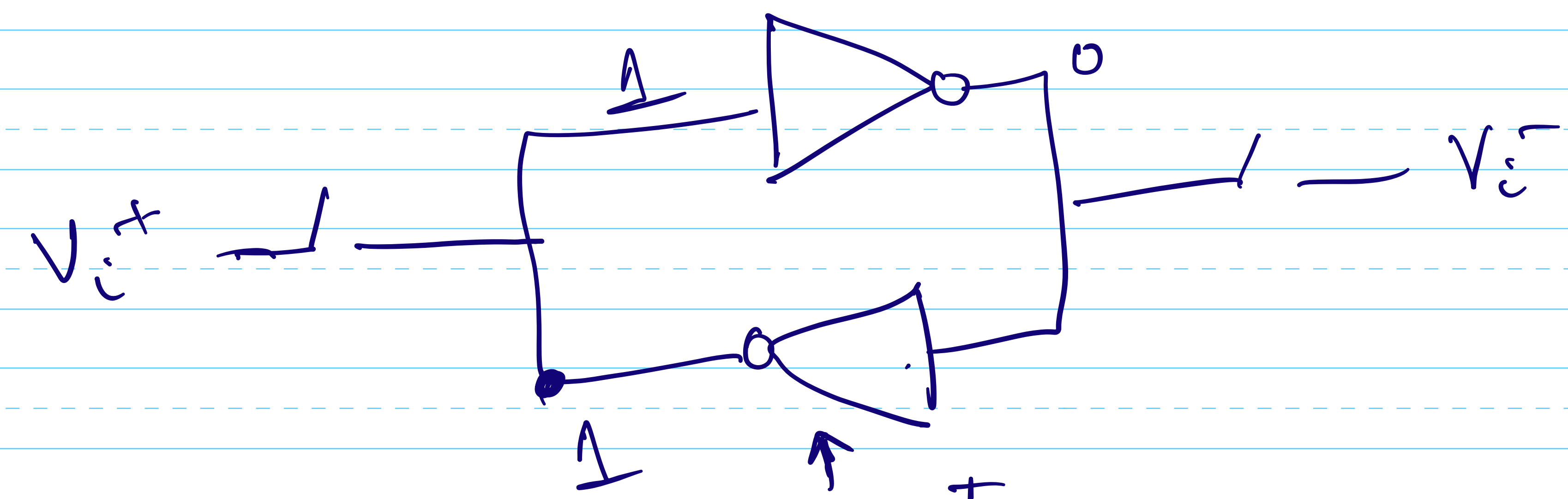
$$\ln(\Delta V) - \boxed{\ln(\Delta V(0))} = \frac{g_m t}{C}$$

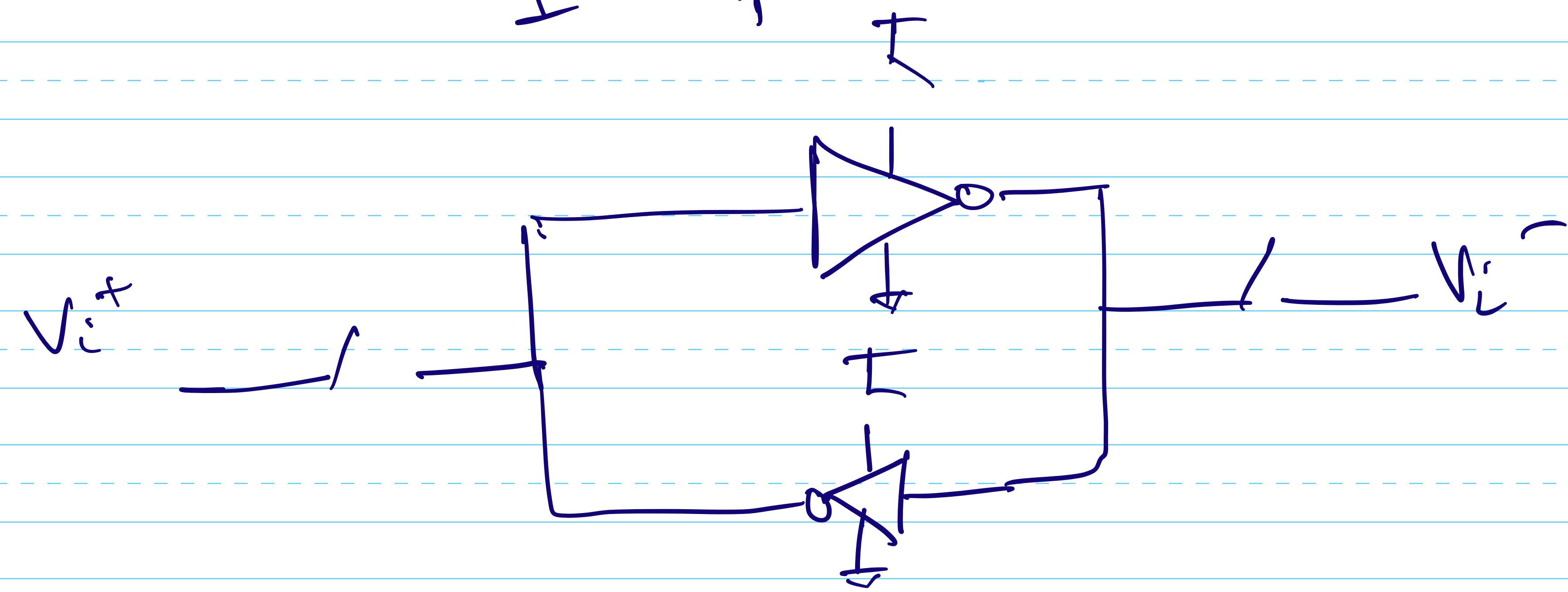
$$\Delta V = \Delta V(0)\, e^{g_m t/C}$$

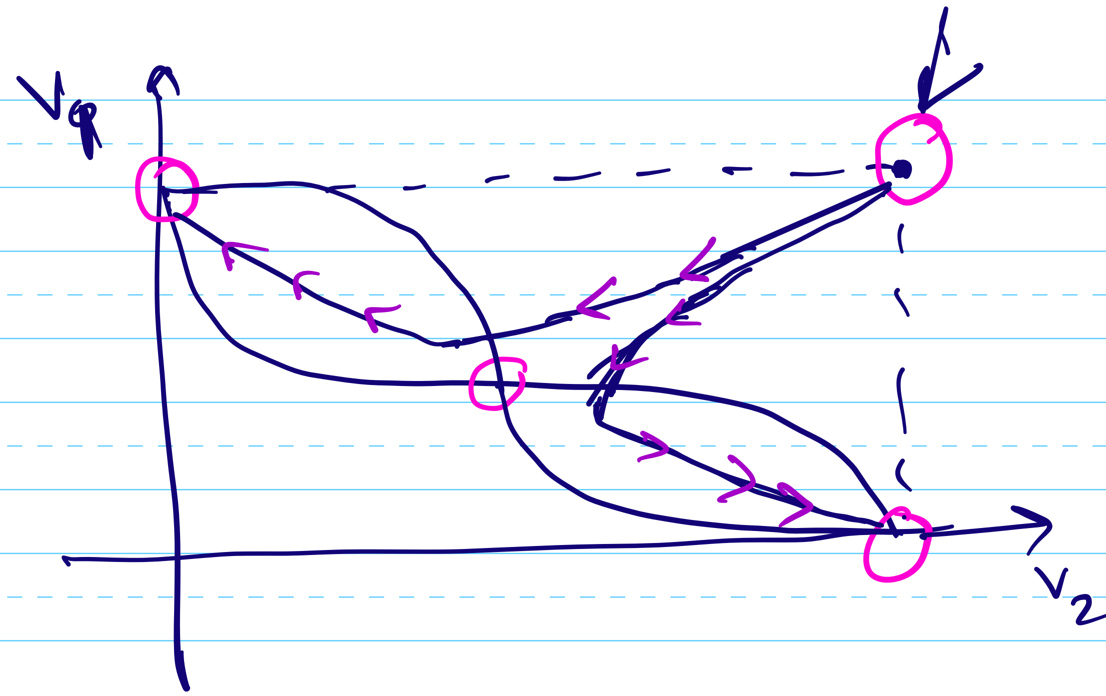


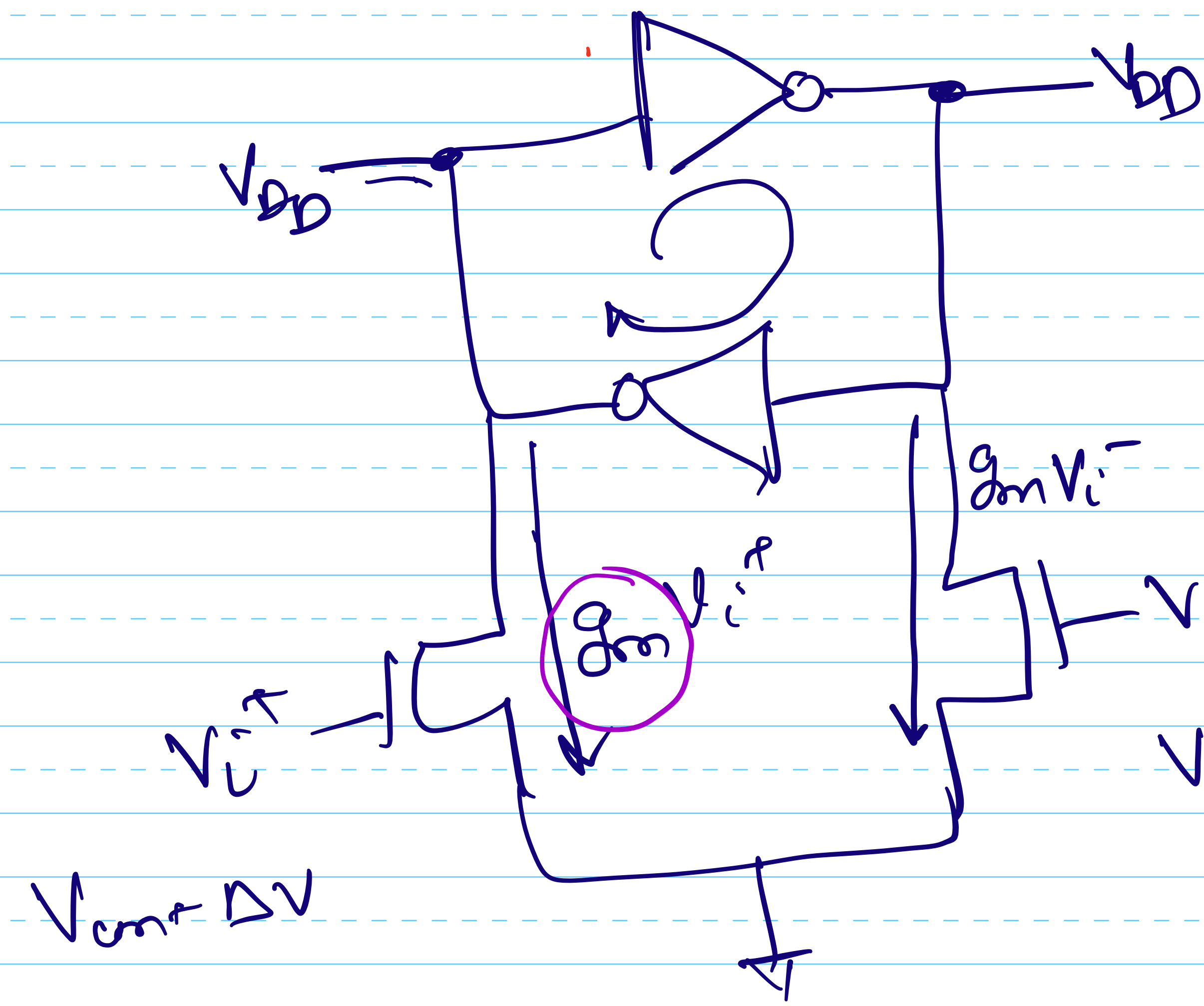


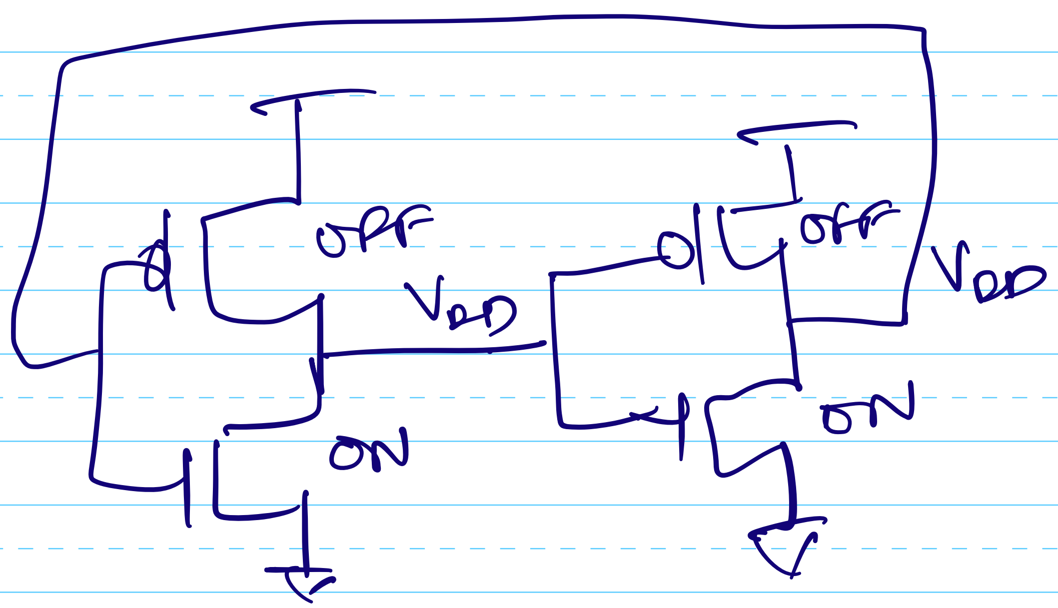
Set up the two inverters in the feedback with the input voltage whose sign needs to be determined.











$$\left[(V_{DD} - V_{tp}) \text{ to } V_{DD} \right]$$

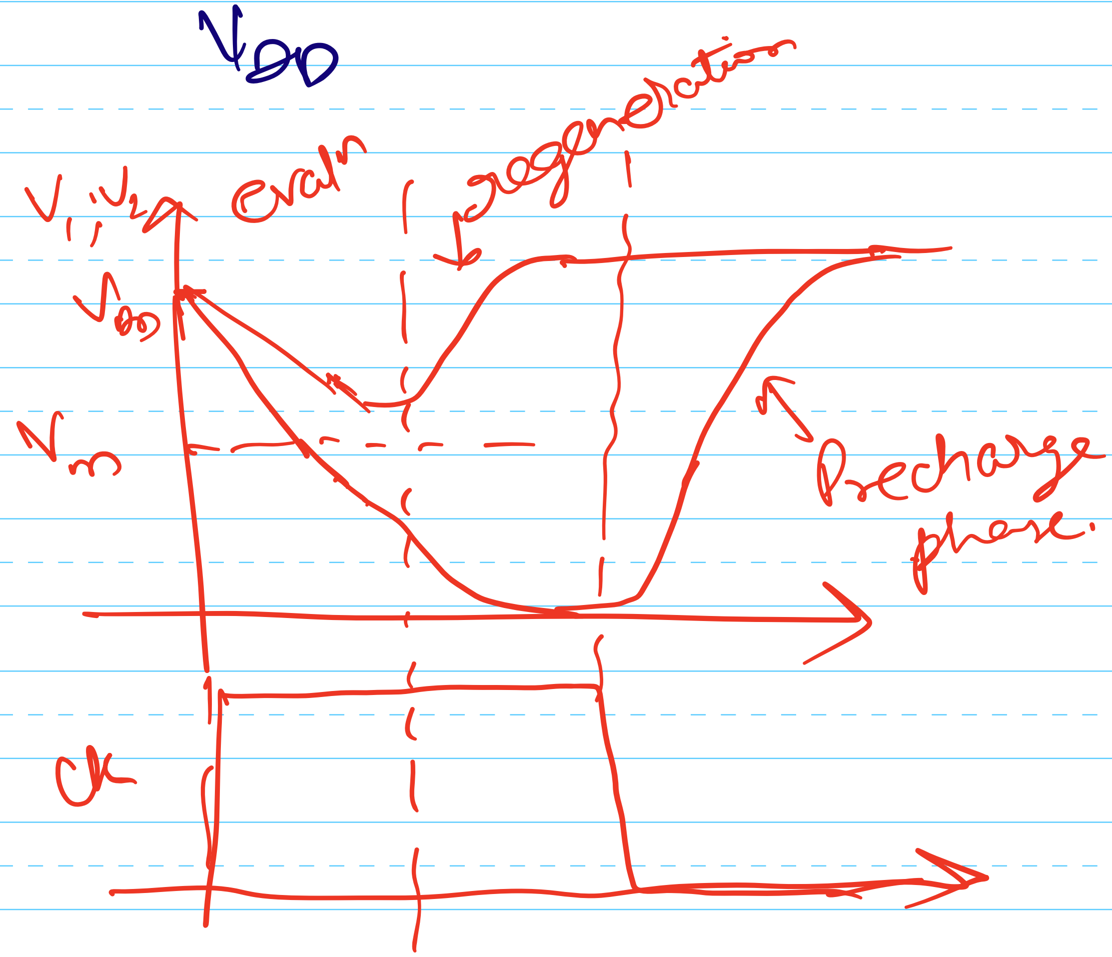
for V_1, V_2 between this range

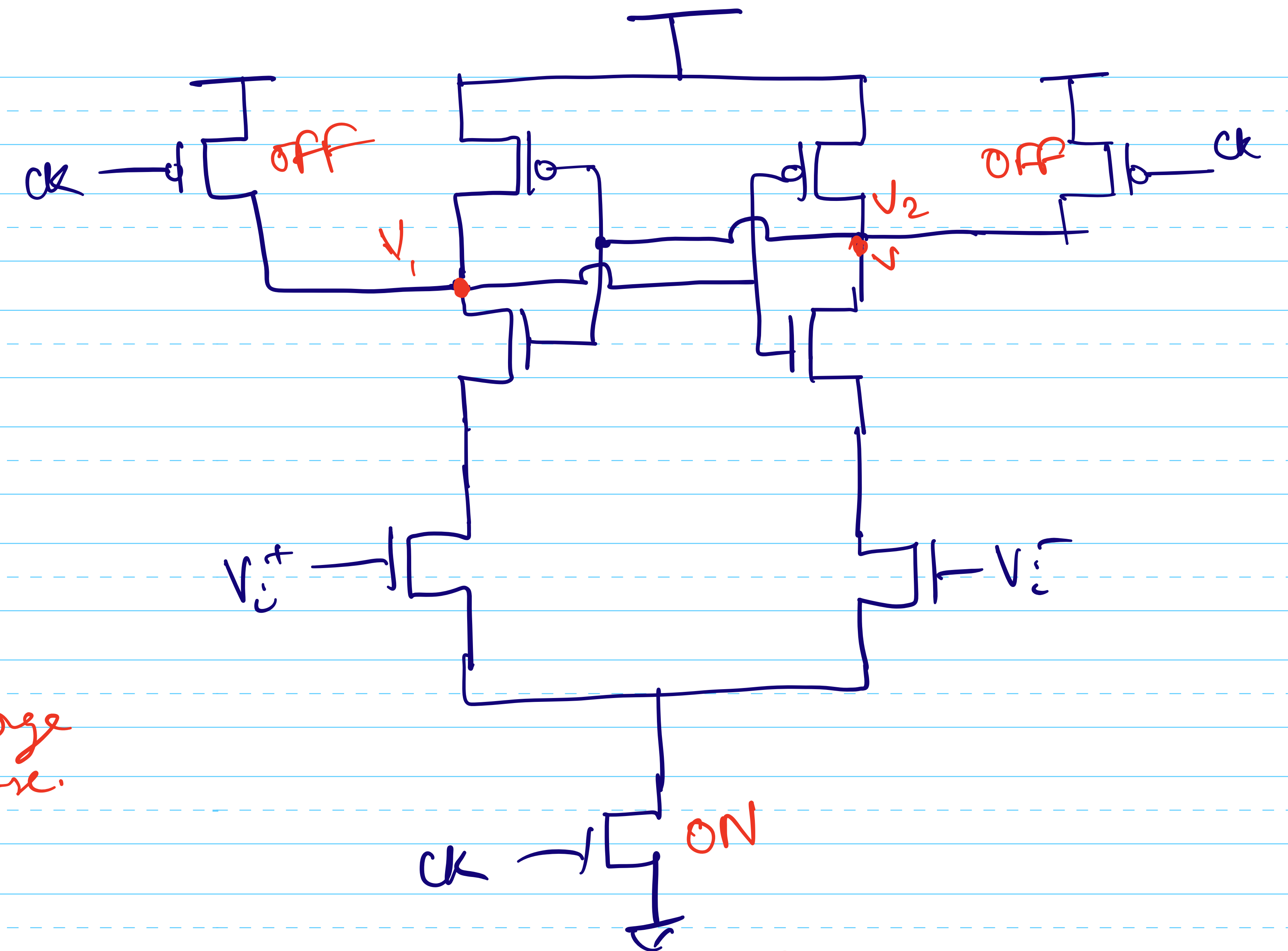
the feedback is not active

When $ck = 0$

O/p → precharged to

V_{DD}





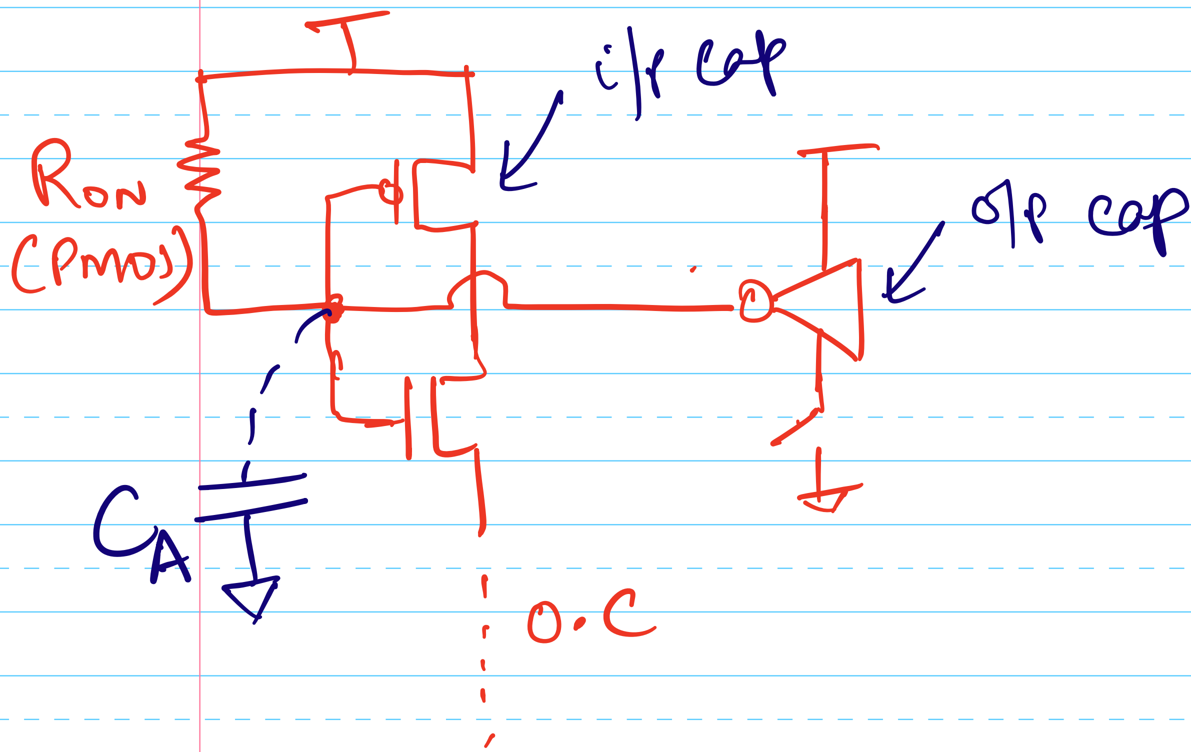
Size of transistors?

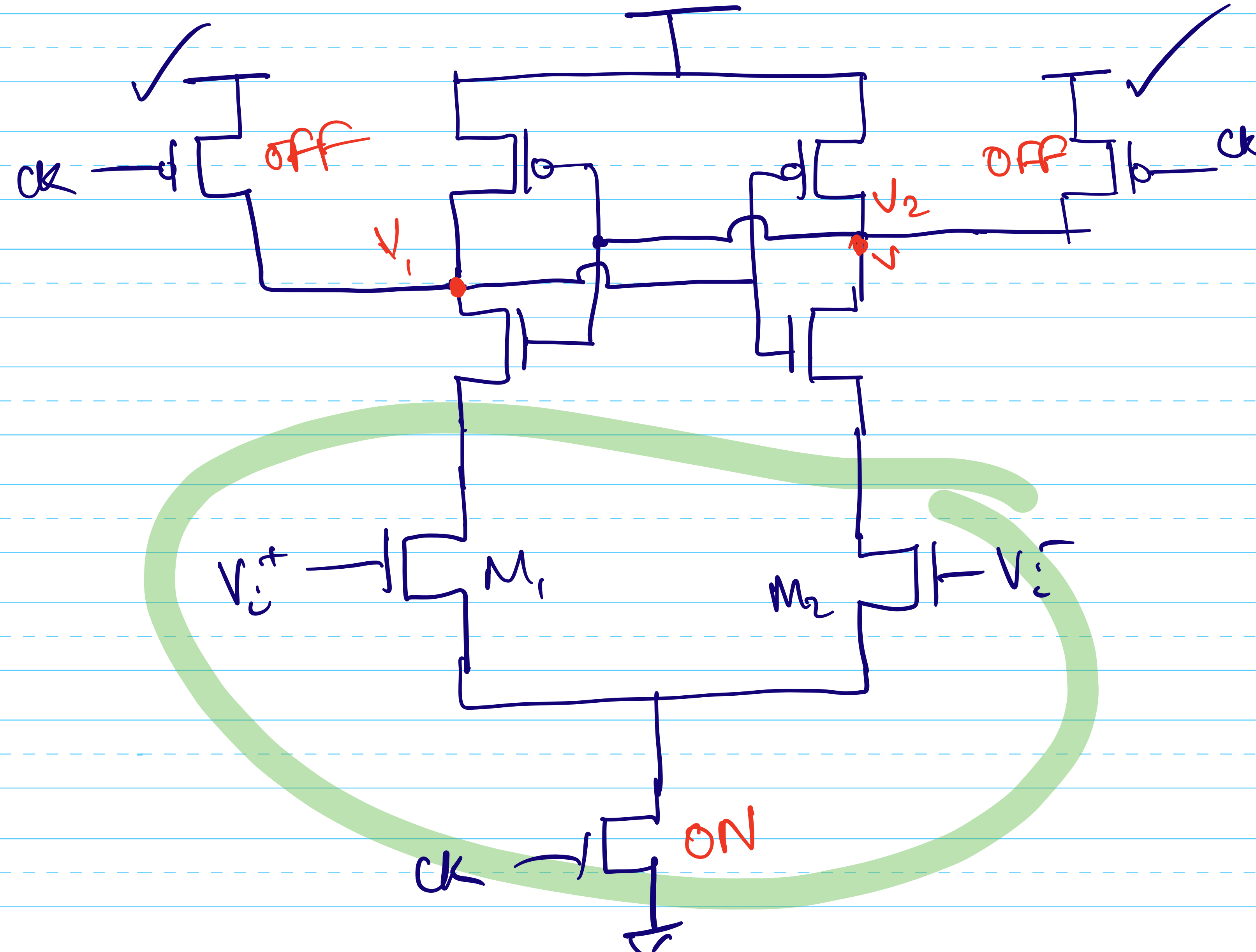
① Precharge phase

② Evaluation phase

③ regeneration

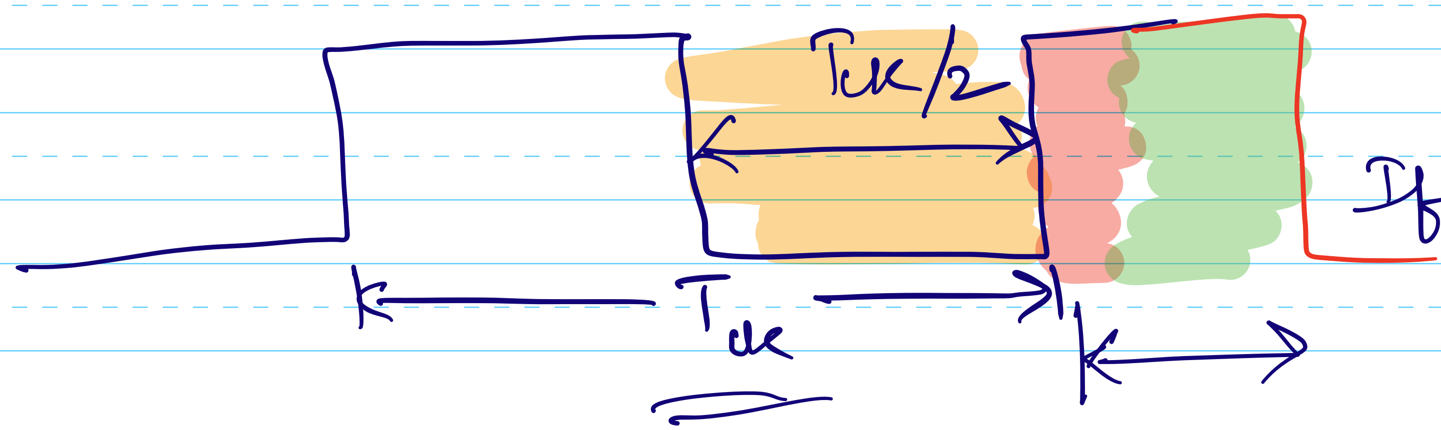
In pre charge phase





PMOS switch size

Size of transistors?

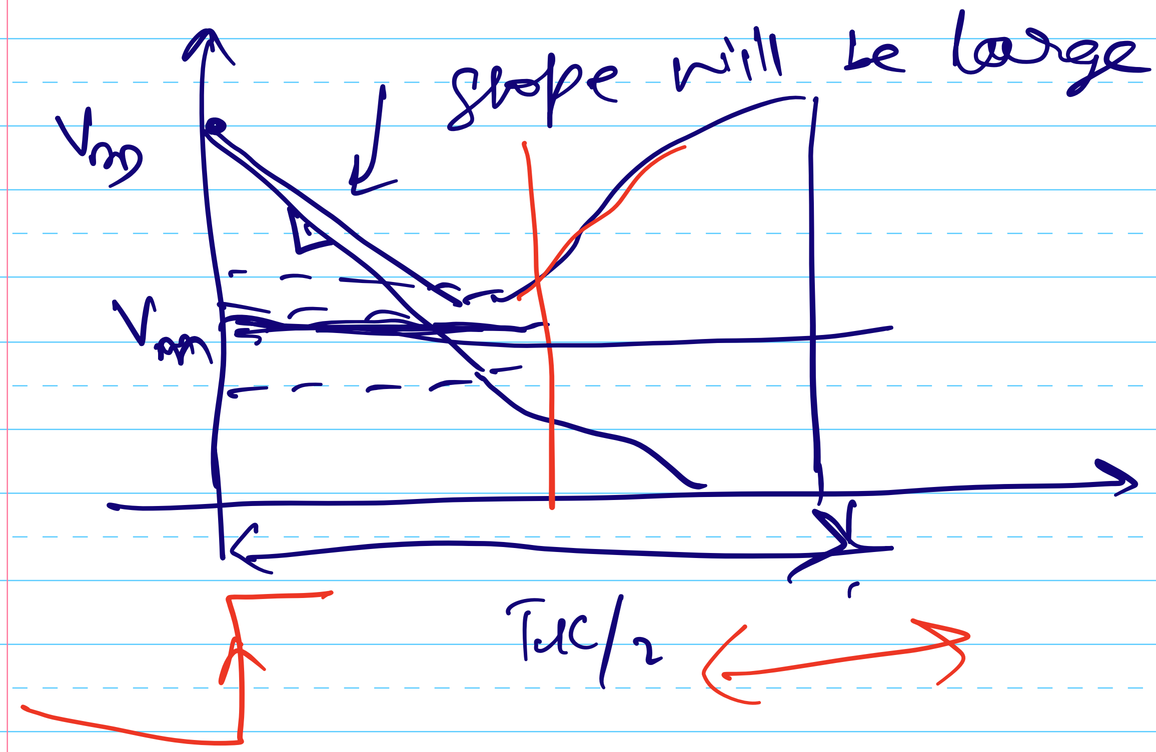


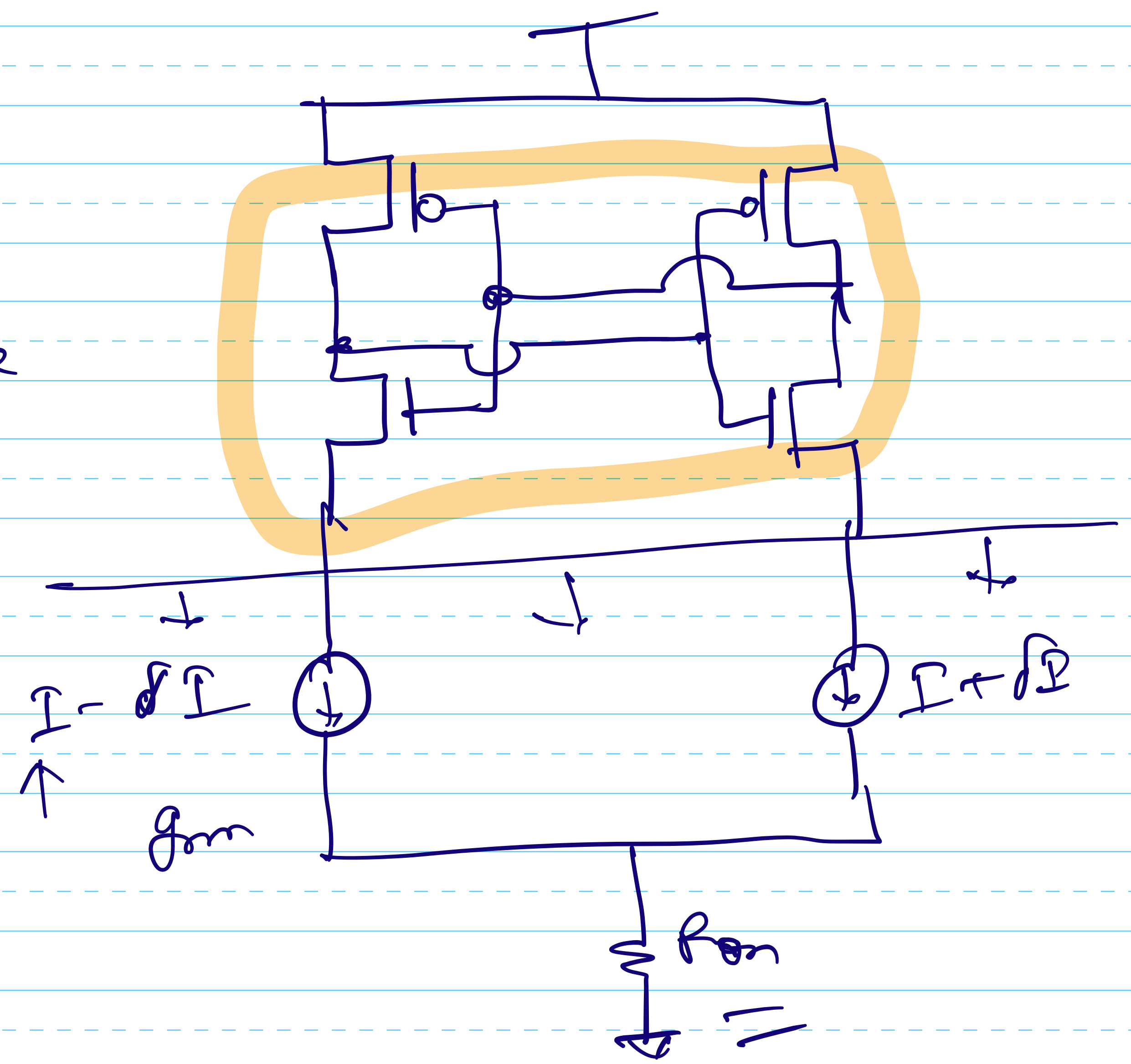
If o/p should charge to $0.99 V_{DD}$

$$R_{on} = \frac{T_{ck}}{9.2\, C_A}$$

Evaluation phase?

I will be large if M_1 & M_2 are large.

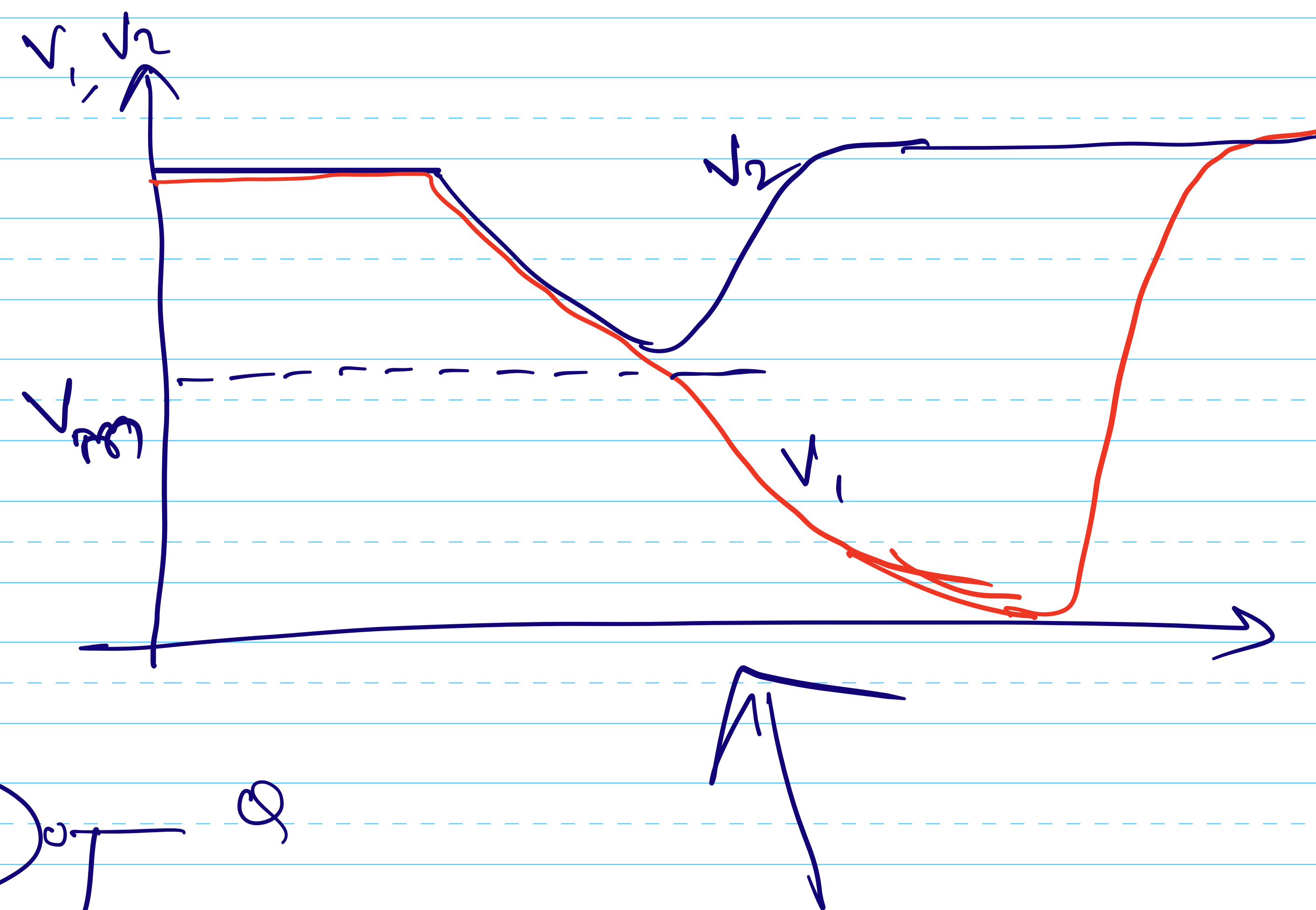


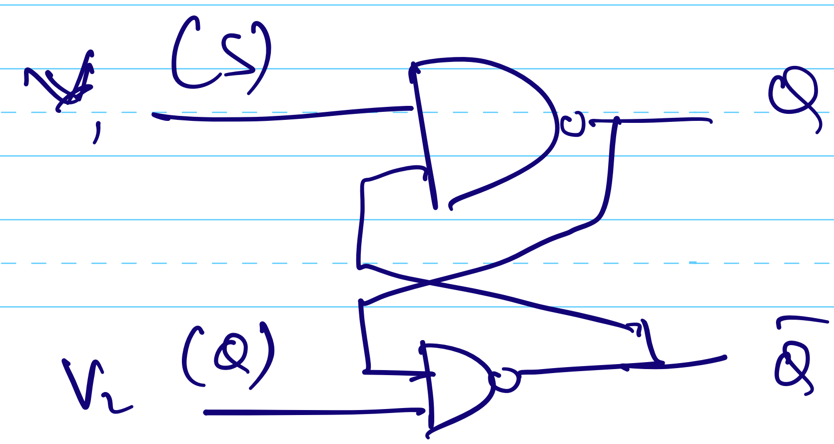


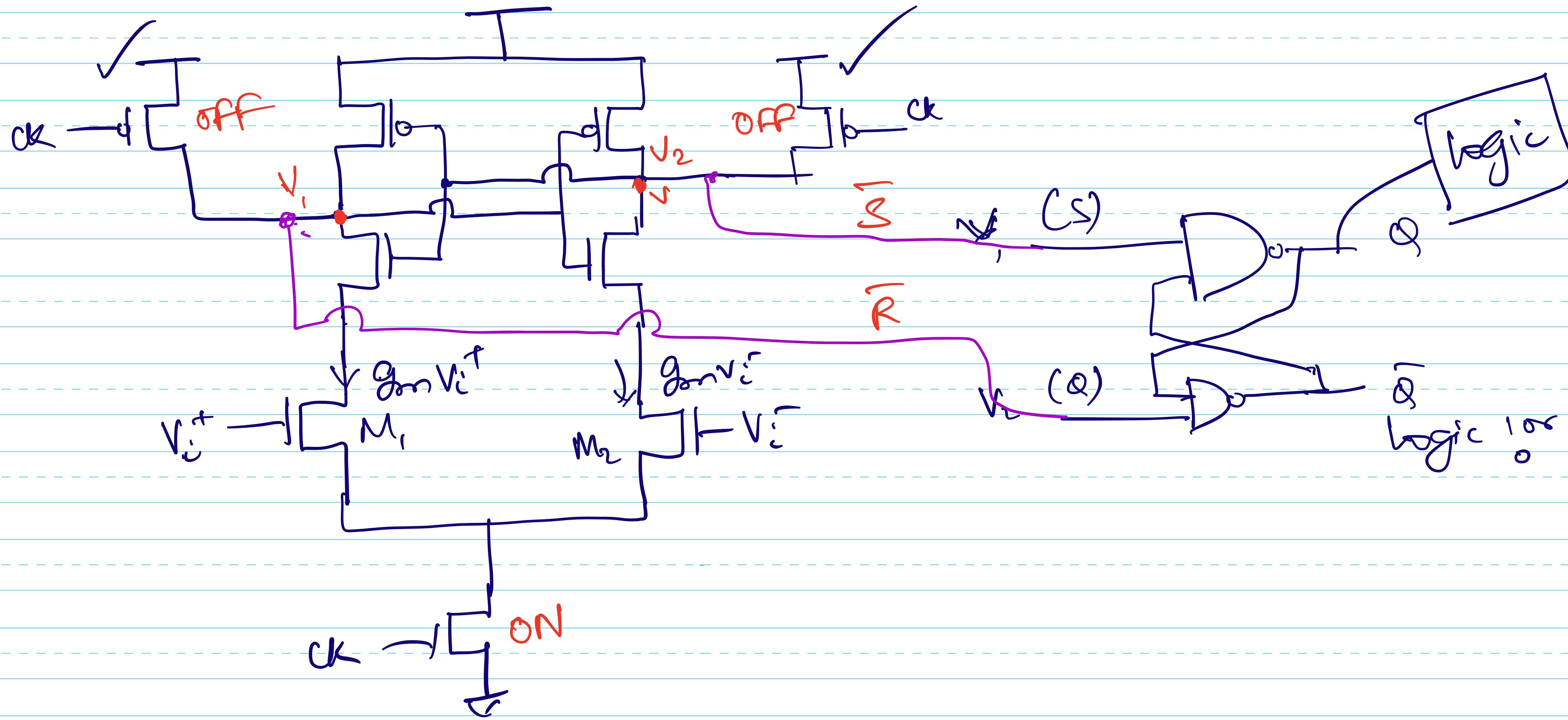
→ M_1, M_2 determine how much time is spent in evaluation phase (along with I/P cm) regeneration phase

Trip point of inverter also controls the time spent in evaluation phase.

O/P of Comparator

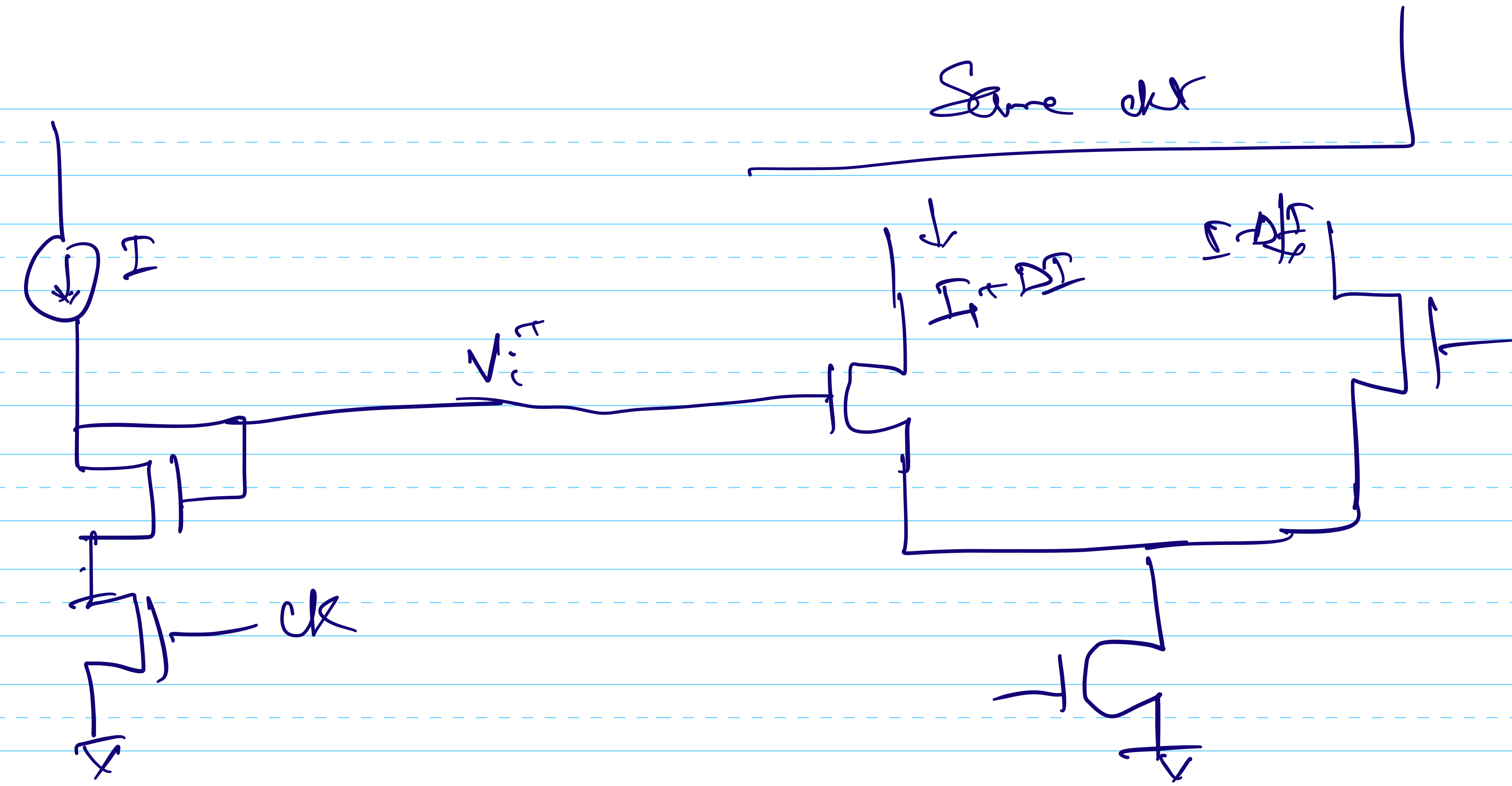




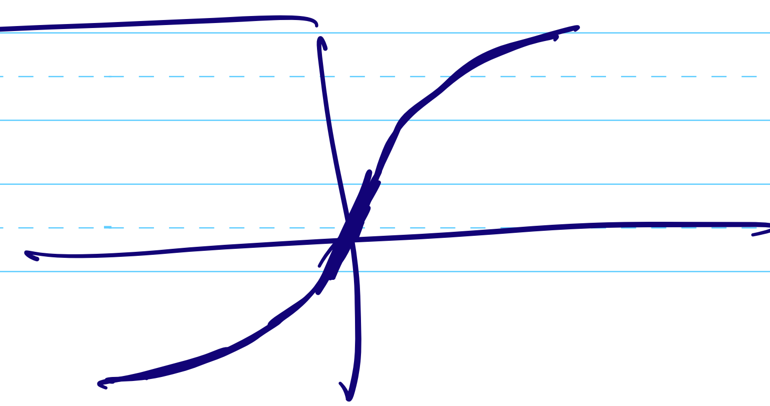


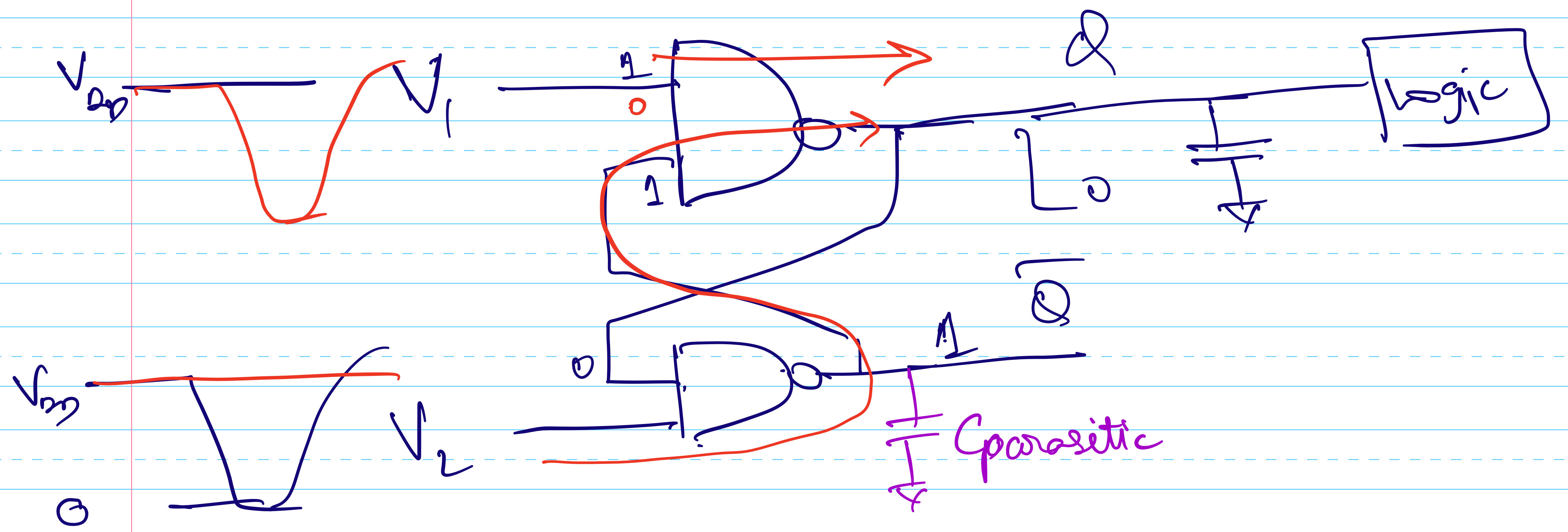
Sense amplifier comparator

Strong ARM Latch comparator

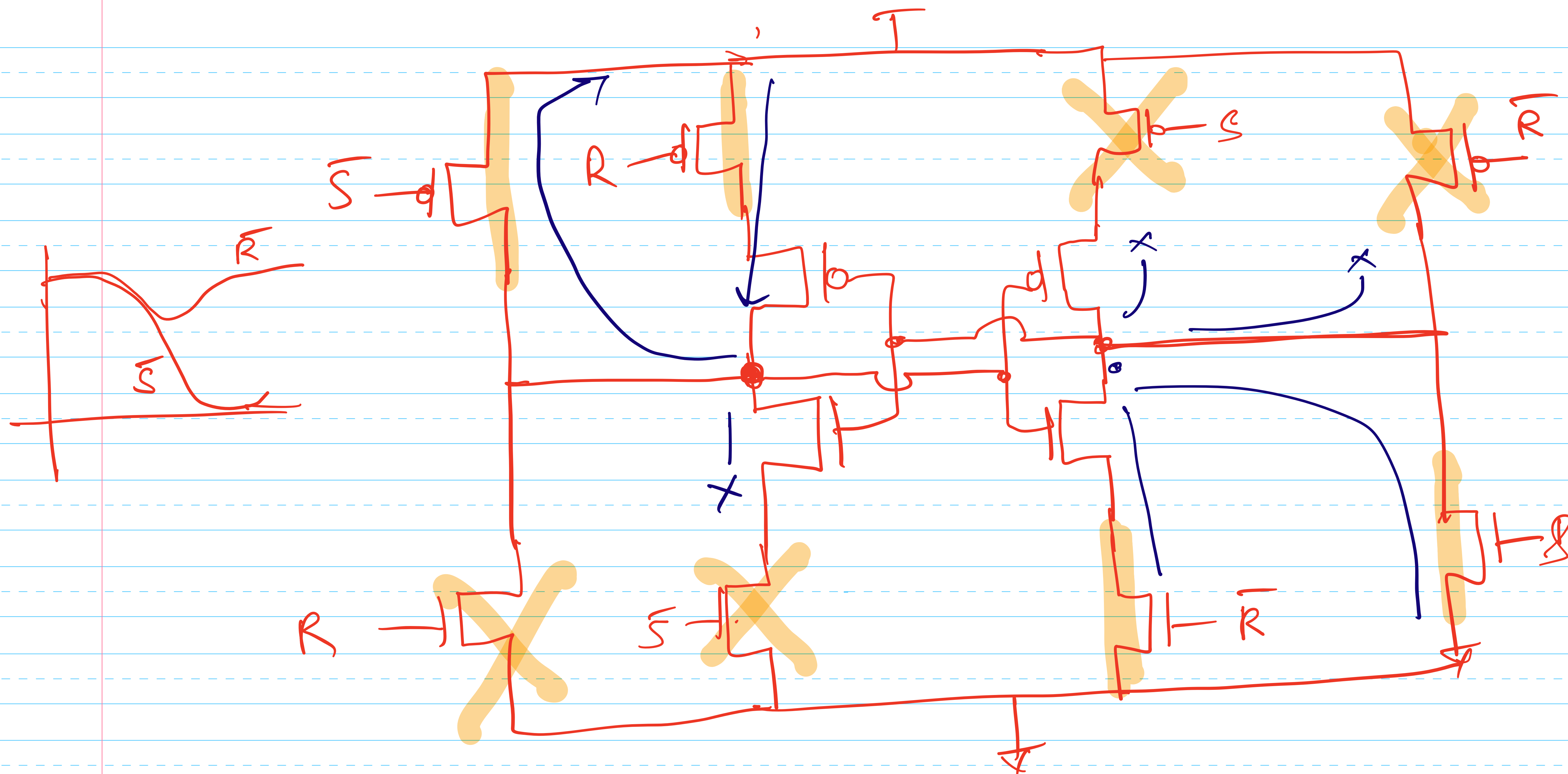


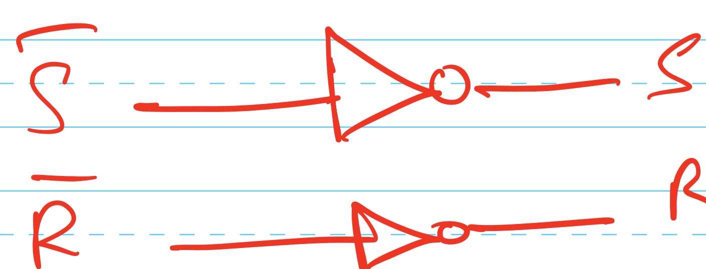
Companding amplifier (log amplifier)





$V_1 \longrightarrow Q$
$V_2 \longrightarrow Q$ } delays will be different





$\bar{S}$ & $\bar{R}$ are at V_{DD} nominally.

Symmetric SR latch